

Companion in the Boundary Problem network. The vehicle the [Hover Disk](#) enables · mechanism in [The Warp Drive](#) · ground-effect sibling The DIY Hoverbike · architecture [How to Build a UFO](#) · first-disk build Viktor Schaubberger §9.

## The Hexalock

Six hover disks in a dodecahedron — building Terrence Howard's six-pentagon collapsed-dodecahedron 6DOF airframe with a  $x_v$  hover disk in each pentagon instead of a propeller. A propellantless, moving-part-free, altitude-independent omnidirectional craft, mixed and driven by one RP2350. The vehicle the hover disk delivers the moment it first locks on the bench.

Osika & collaborators — draft 0.3, 2026-06-20.

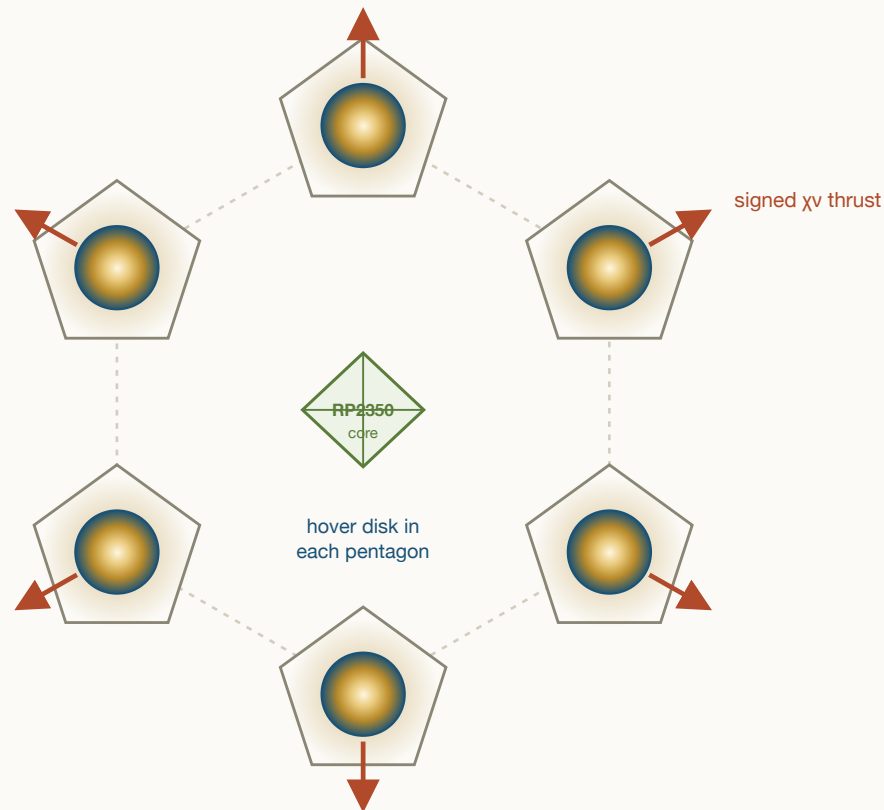


Figure 1. The Hexalock, schematic (not to scale). Six  $x_v$  hover disks — one in each pentagon of the collapsed-dodecahedron shell (dashed) — give six independent, electronically-signed thrust vectors about a central octahedral avionics + power core driven by one RP2350. No propellers; the first-article disks are the spun charged copper shells of §5.1.

**Naming.** The airframe is the published **Lynchpin** geometry; the  $x_v$ -disk craft built on it is what this paper calls the **Hexalock** — six locking disks, one lattice cell.

The **Lynchpin** (Howard, Molter, Seely & Yee, SAE Int. J. Aerosp. 2023) is a true six-degrees-of-freedom airframe: a collapsed dodecahedron of six pentagons, one vectored thruster per pentagon, giving six linearly independent thrust vectors and a square, invertible mixing matrix — so it translates and rotates independently ("tangential flight": hold position at any attitude, or slide any direction while level). The flown prototype put a propeller in each pentagon. **This paper puts a [hover disk](#) in each pentagon instead.** A hover disk does not lift the way a rotor does — it couples to the  $x_v$ /vertex-sharing channel, warping the spacetime lattice so the lattice itself holds the craft. To hover is to **lock**: engage the coupling and the craft is pinned to its pose the way a book is held by a table or a maglev car by its rail ([warp\\_drive.html](#) §2) — the support equals the weight, but it is a static reaction carried by the lattice, not a pumped momentum flux, so it does no work ( $P = F \cdot v = 0$ ) and costs  $\approx$  **zero power**. To move is to **thrust**: modulate the coupling asymmetrically and the same disks apply a net vectored force,  $P_w = (S_{coh} f_{cons}) \cdot \kappa \cdot \frac{1}{2} \epsilon_r \epsilon_0 E^2$ , magnitude and sign on the drive's amplitude and asymmetry  $\kappa$ . That makes the disk the ideal 6DOF actuator: signed thrust with **no propeller, no spinning mass, no reversal latency, no air** — and a hover that is a lock, not a continuous lift. Six disks  $\rightarrow$  six signed  $x_v$  thrust vectors  $\rightarrow$  the same flight-proven 6DOF mixer, now propellantless and vacuum-capable. One **RP2350** runs it: its twelve PIO state machines generate the six disks' drive (amplitude +  $\kappa$ -steer) phase-locked, the two Cortex-M33 cores run the quaternion SE(3) mixer and the estimator, and the firmware is the same

[12\\_PORT\\_SUBSTRATE\\_ADAPTER](#) the optical  $x_v$  chambers and the hoverbike already share. **Honest status:** this is the vehicle the hover-disk program delivers — the geometry and the 6DOF  $x_v$  mixing are sound, buildable reasoning (and the airframe's 6DOF control is flight-proven with the propeller predecessor), but the coupling itself inherits the hover disk's open status: the force law is derived in form, its magnitude is unmeasured and null to date. **This paper also documents the practically-simple first-disk build** — the Repulsine-derived spun charged copper shell of [viktor\\_schauberger.html](#) §9 (§5.1), the cheap first article for the bench, not the diamond ideal. Build the single disk that locks — self-supports its own weight — first ([hover\\_disk.html](#) bench); this is what six of them become.

**Companion papers.** The thruster: [hover\\_disk.html](#) (the  $x_v$  lift body — diamond/SiC/graphene substrate ladder, the four ethers, the drive/capacitor architecture). The mechanism: [warp\\_drive.html](#) (propellantless lattice drive; §3 the force-as-pressure law, §2 why hover is  $\approx$ free, §13 the one coherence dial). Conventional sibling: [diy\\_hoverbike\\_paper.html](#) (the RP2350 / "engineering-optimum" template). Architecture: [how\\_to\\_build\\_a\\_ufo.html](#) (rim-segmented coherent-shell / saucer morphology the disk reuses). First-disk build (Architecture C): [viktor\\_schauberger.html](#) §9 (Repulsine + Tesla). Force law: [CHI\\_NU\\_FORCE\\_LAW.md](#). Root ontology: [shape\\_of\\_the\\_universe.html](#) (the {5,3,3} dodecahedral lattice this drone is a cell of). Geometry & flight-proven 6DOF predecessor: Howard et al., SAE 2023, doi:10.4271/01-16-03-0018.

## §1 What this is — the vehicle the hover disk enables

The opening premise of this paper is the closing promise of *The Hover Disk*: *once a single disk first locks — self-supports its own weight — on the bench, what do you build with six of them?* The answer is a Lynchpin — because the Lynchpin geometry is the one airframe that turns six independent  $x_v$  thrusters into a fully-controllable craft. Each pentagon face carries a hover disk; the six disks point along six independent axes; their  $x_v$  thrusts span the full space of forces and torques. The result is a propellantless, silent, moving-part-free, altitude-independent omnidirectional vehicle.

This is deliberately **not** the propeller drone. The flown SAE prototype used bidirectional propellers, and its own headline limitation was the latency of reversing them; the current public program moved to collective-pitch props to fight the same latency. We skip that whole branch. A hover disk has *no* rotating mass to reverse — its thrust sign is a property of the drive field's asymmetry, set in microseconds with no mechanical state — so it is not an improvement on the propeller, it is a different category of actuator. The propeller version's only role in this paper is as a de-risking *predecessor*: it proves the six-pentagon 6DOF mixing flies with real vectored thrusters (§3). We keep its flight-proven control architecture and replace the thruster.

This paper covers:

- the Lynchpin geometry and why it is also the project's spacetime cell (§2);
- the flight-proven 6DOF mixing the disks inherit (§3);
- why a  $x_v$  disk is the optimal 6DOF actuator — signed thrust, no parts, free hover (§4);
- the **practically-simple first hover-disk build** — Architecture C, the cheap spun charged copper disk to put on the bench (§5.1);
- the full RP2350 build: the six disk thrusters, the drive, the controller, docking (§5–§6);
- the field guide of what breaks, and the honest, bench-conditional status (§7, §9);
- the one coherence dial that runs this same airframe from drone to warp-craft (§8).

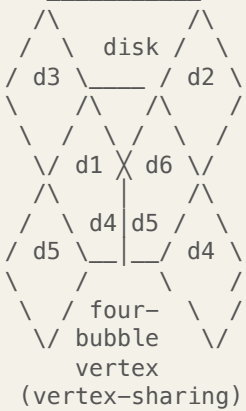
**Open — to be confirmed on the bench** · The load-bearing honesty, stated once, up front. The hover disk's lift is the *open* question of the companion papers: the force law  $F = (C_{\text{sub}} |x_v|^2) \cdot U_{\text{drive}} \kappa / L_{\text{grad}}$  is derived in *form*, but its *magnitude* is unmeasured and null on every substrate tested to date. This paper does not claim the disk lifts. It specifies the craft you assemble *when it does* — and shows that the geometry, the 6DOF mixing, and the RP2350 control are sound and buildable reasoning regardless, so the only open input is the one number the `warp_drive.html` §7 bench measures: where a real disk sits on the coherence dial.

## §2 The geometry — a collapsed dodecahedron, which is our cell

The Lynchpin airframe is, in the SAE paper's own words, "a collapsed dodecahedron" — an icosi-dodecahedral / vector-equilibrium form built from **six pentagons** of equal edge length  $a$ , whose six

planes are *all* mutual symmetry planes, derived from "four Tetryen elements" of Jeff Yee's energy-wave particle model. Each pentagon presents one face normal; here, each carries one hover disk, and the six normals are the craft's six thrust axes.

the six-pentagon "collapsed dodecahedron"  
(all 6 pentagon planes are symmetry planes)



6 face-normals  $\Rightarrow$   
6 lin. indep.  
 $\chi_v$  thrust vectors  
 $\Rightarrow$  6x6 mixing M  
 $\Rightarrow$  full SE(3) / 6DOF

"tangential flight":  
translate & rotate  
independently  
(orbit own CoM;  
hover at any attitude;  
hold  $\approx$  free power)

one pentagon = one hover disk  
 $\chi_v$  thruster on the face normal  $\hat{n}_i$   
signed pressure  $P_i \in [-P_{max}, +P_{max}]$

$\hat{n}_i$   
 $\uparrow$  (lattice momentum sink)



no rotor  $\cdot$  no air  $\cdot$  no latency

central octahedral  
avionics + power core  
(nested Platonic solid)

*Schematic, not to scale. Six pentagons  $\rightarrow$  six signed  $x_v$  thrust vectors  $\rightarrow$  a square invertible mixer  $\rightarrow$  genuine six-degrees-of-freedom flight, propellantless. The vertex where pentagons meet is the "four-bubble" / four-Tetryen junction — in the companion frame, a vertex-sharing site, which is the very channel each disk locks onto.*

Here is the part that is not a coincidence to anyone who has read the root papers. The whole Boundary-Problem network rests on one geometric claim: spacetime is the **{5,3,3} dodecahedral honeycomb**, whose primitive cell is the pentagon-faced dodecahedron (the carbon dodecahedrane C<sub>20</sub> is its literal molecular realization; [hover\\_disk.html](#) §13). The Lynchpin's airframe *is that cell* — chosen, independently, by aerodynamicists optimizing a thrust frame — and a hover disk holds precisely *because* its body resonates with that lattice cell (the consonance ether, [warp\\_drive.html](#) §13). So putting hover disks in a Hexalock is not bolting one program onto another: it is building the lattice cell out of six smaller lattice-cell resonators. The shape that makes the best omnidirectional thrust frame is the same shape that makes the best lattice coupling.

| Lynchpin term (SAE 2023 / Tangential Flight)   | Network term                                                                         | Same object?                                                                                                                                  |
|------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| collapsed dodecahedron / six pentagons         | {5,3,3} primitive cell / dodecahedrane C <sub>20</sub>                               | ESTABLISHED geometry; identification exact (same solid)                                                                                       |
| "four Tetryen elements" / four bubbles meeting | vertex-sharing site ( $g \leq 1/\sqrt{3}$ , <a href="#">G_VERTEX_DERIVATION.md</a> ) | FRAMEWORK INFERENCE — the $x_v$ lift channel <i>is</i> vertex-sharing; the four-fold (tetrahedral, 109.47°) junction is a Plateau/foam vertex |

| Lynchpin term (SAE 2023 / Tangential Flight)   | Network term                                                                  | Same object?                                                                                   |
|------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| "electromagnetic vacuum surfaces" (docking)    | $x_v$ coupling surface (the lattice momentum sink)                            | FRAMEWORK INFERENCE — §8; Howard's docking face is the disk's lift face                        |
| super-symmetric, fractal, "scales to any size" | size-independent self-lift $F/W \propto R$ ( <code>hover_disk.html</code> §5) | DERIVED — $x_v$ self-lift is a material property, not a size; the airframe scales the same way |
| central octahedral avionics core               | nested Platonic solid (octahedron $\subset$ dodecahedron)                     | ESTABLISHED — a real choice in the prototype, kept here                                        |

**Doctrine sentence** · *The Lynchpin airframe is the spacetime-lattice cell built as an aircraft, and the hover disk is the lattice cell built as a thruster. Assembling six disks into a Hexalock builds the cell out of cells — the same dodecahedral resonance at two scales. That is why this airframe and this thruster were made for each other: they are the same geometry, nested.*

### §3 Tangential flight — six signed thrust vectors (flight-proven)

In an ordinary multirotor, translation and rotation are *coupled*: to fly forward you pitch forward, because all rotors point up. The Lynchpin's six thrusters point along six different axes, so their thrusts *span* the space of forces and torques. Collect the six unit thrust directions as the columns of a  $3 \times 6$  matrix and the six moment arms likewise; stack them into the  $6 \times 6$  map  $M$  from per-disk thrusts to body wrench:

$$\begin{bmatrix} \mathbf{F} \\ \mathbf{M} \end{bmatrix} = M \begin{bmatrix} P_1 \\ \vdots \\ P_6 \end{bmatrix}, \quad \begin{bmatrix} P_1 \\ \vdots \\ P_6 \end{bmatrix} = M^{-1} \begin{bmatrix} \mathbf{F} \\ \mathbf{M} \end{bmatrix}$$

Because the six axes are linearly independent,  $M$  is square and non-singular: it **always inverts**. Name any wrench — a force in any direction, a torque about any axis, the two decoupled — and  $M^{-1}$  hands you the six thrusts that produce it (SAE 2023, Eqs. 8–10). That is the whole of "tangential flight": translate while holding attitude (stare a camera at a fixed point while orbiting it), rotate while holding position (hover upside-down, pivot in place), or both at once — the full SE(3) envelope from six thrusters.

**Derived / established** · This half is flight-proven. Howard, Molter (University of Stuttgart), Seely & Yee designed, built, and *flight-tested* the six-pentagon mixer with real vectored thrusters (a 651 g

propeller prototype, Pixhawk 4 Mini / PX4, quaternion control), peer-reviewed in the *SAE Int. J. Aerosp.* (2023). So the geometry, the invertible mixer, and the quaternion 6DOF control law are not in question — they flew. **Swapping the thruster type from propeller to hover disk does not touch the mixer**;  $M^{-1}$  is the same matrix whether the six  $P_i$  are propeller thrusts or  $x_v$  pressures. We inherit a working control stack and change only the actuator physics.

**Implementation consequence · The disk dissolves the prototype's one hard problem.** A pure six-thruster 6DOF vehicle must produce *signed*, arbitrarily-small thrust on each axis (the near-zero-thrust constraint that pushed ETH's Omnicopter to eight over-actuated rotors, and forced the Lynchpin prototype into bidirectional props with their reversal latency). A hover disk meets that requirement for free: its thrust is  $P_w \propto \kappa \cdot U_{\text{drive}}$ , continuous through zero, and its sign is the sign of the drive asymmetry  $\kappa$  — no spinning mass passes through zero, so there is no reversal transient at all (§4). The geometry's hardest demand is exactly the  $x_v$  actuator's natural behaviour.

One structural fact the prototype surfaced and we keep: unlike a normal multirotor, the **z-position of the centre of gravity changes the thrust solution** (the six axes are not coplanar). Keep the heavy mass — power and drive electronics — on the geometric centre in the octahedral core, and fold any residual offset into  $M$  at arm time, as the prototype did with its 42 mm battery offset.

## §4 Lock, not lift — why a $x_v$ disk is the optimal 6DOF actuator

First, the word, because it changes how the whole craft reads. A propeller drone *lifts*: it runs a pump that throws air downward every second, and the upward force is the reaction to that endless momentum flux — cut the power and it falls. A  $x_v$  craft does not lift. When it holds station it **locks**: it engages the vertex-sharing coupling and the spacetime lattice *holds its pose*, the way a book is held by a table or a maglev car by its rail ( [warp\\_drive.html](#) §2, the "Omicron" hover). The support still equals the weight, but it is a *static reaction carried by the lattice*, not a pumped flux, so it does no work ( $P = \mathbf{F} \cdot \mathbf{v} = 0$ ) and costs  $\approx$  zero power. To *move*, the craft modulates the coupling asymmetrically and the same disks apply a net vectored **thrust** (warp §2's "Delta" / travel) — which does cost work, recoverable regenerative-braking style. The Lynchpin has those two regimes, and "lift" names neither:

| Mode                               | What it is                                                                                                                | Power                                                                |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Lift (PROPELLER — THE WRONG MODEL) | a continuous downward momentum pump; the reaction holds you up                                                            | continuous — stop and you fall                                       |
| Lock ( $x_v$ HOVER)                | engage the coupling; the lattice pins the full pose — position <i>and</i> attitude (book on a table / maglev in its rail) | $\approx 0$ — no work is done; $P = \mathbf{F} \cdot \mathbf{v} = 0$ |



| Mode                            | What it is                                                                          | Power                                   |
|---------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------|
| <b>Thrust</b> ( $x_v$ MANEUVER) | modulate the coupling asymmetrically → a net force/torque to <i>change</i> the pose | costs work — recoverable (regenerative) |

With that fixed, the engineering-optimum argument is not "which propeller" — it is that the propeller is the wrong category of device for this airframe, and the hover disk is the right one. Lay the two against the requirements a 6DOF craft actually has:

| Requirement                            | Propeller (any pitch scheme)                                   | $x_v$ hover disk                                                                                                                            |
|----------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Signed thrust, continuous through zero | needs reversal — motor spin-down/up or pitch slew through flat | $\kappa$ sign flips the gradient; smooth through zero, no transient                                                                         |
| Reversal / response latency            | the prototype's stated bottleneck (tens of ms / servo-limited) | field-set; microseconds, no mechanical state to change                                                                                      |
| Moving parts per thruster              | rotor, bearings, (servo, linkage)                              | none in the materials-ideal disk; one constant-RPM flywheel in the simple Architecture-C first build (§5.1) — never a thrust-reversing part |
| Works without air                      | no — thrust → 0 with density; ceiling ~30 km                   | yes — the hold is a lattice lock, altitude-independent                                                                                      |
| Power to <i>lock</i> a hover           | continuous (100s of W) — pumping against gravity every second  | $\approx 0$ — the lattice holds the pose; $P = F \cdot v = 0$ (warp §2)                                                                     |
| Acoustic signature                     | broadband rotor noise                                          | silent (no aerodynamic source)                                                                                                              |
| Thrust direction within a face         | fixed along the shaft                                          | steerable — rim-segment $\kappa$ can vector within the disk (bonus authority)                                                               |
| Coupling magnitude — THE CATCH         | known, available today                                         | OPEN — null to date; the bench question (§9)                                                                                                |

The disk wins every row but the last, and the last is the whole game. The disk's thrust is the warp pressure of [warp\\_drive.html](#) §3, written per face:

$$P_w = (S_{\text{coh}} f_{\text{cons}}) \cdot \kappa \cdot \frac{1}{2} \varepsilon_r \varepsilon_0 E^2, \quad \text{lock condition } P_w \geq \rho t g$$

with  $S_{\text{coh}}$  the coherent fraction (life ether),  $f_{\text{cons}}$  the consonance fidelity (chemical ether, ceiling  $e^{-P/12} \approx 0.873$ ),  $\kappa \in [-1, 1]$  the drive asymmetry that sets the vector's *magnitude and sign*, and  $\frac{1}{2} \varepsilon_r \varepsilon_0 E^2$

the stored field-energy density ( $\approx 25$  MPa for a diamond stack at 1 GV/m). A disk reaches the lock condition — supports its own weight — when its drive pressure exceeds its areal weight  $\rho t g$  — a *size-independent* condition (the area cancels), so it is a property of the disk's *material and drive*, not its diameter. The warp paper's §13 dial says the coupling  $S_{\text{coh}} f_{\text{cons}}$  need only clear  $\sim 10^{-6}$  for a thin diamond skin to carry its own weight — a sliver of full coherence.

**Implementation consequence** · **Thrust is a number you write to a register.** Each disk's commanded thrust  $P_i$  maps to a drive setpoint ( $|\text{amplitude}| \rightarrow \text{field } E, \text{ sign} \rightarrow \kappa$ ). There is no rotor to govern, no pitch to slew, no thermal/aero lag — the actuator is the drive waveform. That is why the §3 mixer becomes trivial to close:  $M^{-1}[\mathbf{F}; \mathbf{M}]$  lands on six clean, fast, linear, bipolar actuators with a true zero. The hardest control problem in 6DOF flight — authority at zero net thrust — is the disk's easy regime.

**Doctrine sentence** · *A propeller converts shaft power into a momentum flux of air, every second, downward; a hover disk borrows a force from the spacetime lattice and gives it back when it lets go. The first is a pump that must run to hold you up; the second is a foot that stands on the floor. For a craft that must hover at any attitude, in any medium, the disk is not a better pump — it is the end of pumping. The craft does not lift; it locks to hover and thrusts to move.*

## §5 The build — six disks, one RP2350, twelve channels

### 5.1 The disk thruster — start practically simple (Architecture C)

`hover_disk.html` ranks the *best* disk: single-crystal diamond in its optical mode, an icosahedral-quasicrystal register, the full four-ether stack. That is the destination, not the first article. The drone's first disks should be a "**pretty simple**" **weekend build**, in the hoverbike's spirit — the cheapest body that exercises enough of the levers to be worth putting on the `warp_drive.html` §7 force balance. The companion `viktor_schauberger.html` §9 names exactly that build, reasoned from history rather than materials science: **Architecture C — the spun, charged, chiral-vortex copper shell**. It is the first-disk spec we adopt, and the reason this drone paper, not the hover-disk paper, is where the simple build lives: the hover-disk paper optimizes the *material*; here we optimize the *first thing you actually machine*.

Schauberger's Repulsine, with Tesla's bladeless turbine and resonant coil beside it, assembles *four* of the warp drive's actuation levers (`warp_drive.html` §4) in copper, with shop tools and no cleanroom: **spin** (a mechanical  $\kappa$ -axis symmetry-breaker), **stored HV charge** ( $U_{\text{drive}}$ ), a **chiral-spiral field** (the asymmetry  $\kappa$  built into the geometry, not the electronics), and a **closed, cold shell** (coherence). It needs no THz/optical drive and no exotic crystal — a spun, charged copper disk does it. The framework's own honesty argues this is the *right* first try: the published electrostatic-thrust



nulls ( [hover\\_disk.html](#) §15) are all *static* capacitors with no high-frequency mode (no predicate P4), so they cannot even test the  $x_v$  channel — and bulk spin supplies that mode *mechanically*, cheaply, the way no static plate can.

| First-article layer       | Practically-simple pick (per pentagon)                                                                       | Lever it buys (warp \$4)                                                                 |
|---------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Body / shell              | two facing domed copper discs across a corrugated "waveline" gap, ~80–120 mm (the pentagon's inscribed disc) | closed cold shell — the Repulsine geometry                                               |
| Spin                      | one small balanced BLDC spinning the shell at a governed constant RPM (a flywheel, <i>not</i> a propeller)   | spin = the $\kappa$ -axis symmetry-breaker (warp \$4-i)                                  |
| Charge / electrode        | a chiral-spiral electrode etched on the disc faces, charged to a few kV                                      | stores $U_{\text{drive}}$ ; the spiral is $\kappa$ 's direction, no steering electronics |
| HV stage                  | a small per-disk flyback or Cockcroft–Walton, RP2350-gated; optional Tesla-coil-style resonant arm           | delivers/stores $U_{\text{drive}}$ ; the resonant arm adds the P4 high-frequency field   |
| Coupling skirt (optional) | a smooth or textured boundary-layer disc (the Tesla turbine)                                                 | a <i>mechanical</i> $\eta$ coupling channel                                              |
| Cooling                   | passive heatsink / thermal mass; colder is strictly better                                                   | the coherence budget — Schauburger's "cold = living"                                     |

The §3–§4 control contract is met with no exotic part. Per disk the RP2350 commands **thrust magnitude** via the HV charge amplitude (electronic, fast) and **thrust sign / vector** via charge polarity / differential bias across the spiral (electronic); the spin is held at a constant RPM by a slow governor and is a *symmetry-breaker, not a control*. So the mixer still writes {amplitude, sign} to six disks (§5.2) — with the one honest caveat that the simple first disk carries a constant-RPM flywheel (a balanced rotor, no thrust reversal, far below the complexity of the variable-pitch propeller it replaces).

**Open — to be confirmed on the bench** · This is a first bench article, not a working thruster. Build one, put it on the [warp\\_drive.html](#) §7 vacuum force balance, and measure self-lift — against the stored-energy,  $\kappa \rightarrow 0$ , and chamber-variation controls — *before* you build six and bolt them to an airframe. Architecture C is the open build hypothesis of [viktor\\_schauberger.html](#) §9: it exercises four levers the static capacitor does not, which is why it is the right first try — not a guarantee that it lifts. Every  $x_v$  measurement to date is null; this is the cheapest configuration that could change that.

**The upgrade ladder — when the simple disk returns a receipt**

The moment a simple disk shows a non-null force on the balance, climb the `hover_disk.html` material ladder for more coupling per disk, and eventually to the no-moving-parts static endpoint (which trades the spin for a high-frequency optical drive). Same socket in the pentagon, better fill:

| Disk layer                 | Materials-ideal pick                                                                  | Why (which ether)                                                                               |
|----------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Structural / coherent core | single-crystal diamond ( $^{12}\text{C}$ -enriched); SiC for a cheaper larger article | energy ceiling (Warmth, $R = 1084$ ) + coherence (Life); optical-mode $\eta$ in one body        |
| Geometry register          | face etched as a 12-fold photonic quasicrystal                                        | consonance (Chemical) — the icosahedral/ $\varphi$ lattice match, in a true insulator           |
| Electrodes                 | graphene, rim-segmented ( $\approx 12$ segments)                                      | atomically smooth (raises breakdown); the segments are the $\kappa$ -steer (replacing the spin) |
| Coupling option            | thin $\text{BiFeO}_3$ (hBN-buffered)                                                  | coupling (Light) — the multiferroic P2 channel                                                  |
| Drive                      | HV with an RF $\rightarrow$ THz arm (per-segment phase/amplitude)                     | stores $U_{\text{drive}}$ ; exercises the \$4 A/B frequency fork — and removes the moving part  |

## 5.2 Why the RP2350, and the channel map

We keep the project's standard controller for the same reasons the hoverbike does, plus one specific to this airframe:

- **Twelve PIO state machines.** The RP2350 has three PIO blocks  $\times$  four state machines = **12** cycle-exact I/O engines. Six disks, each driven by two channels (amplitude gate +  $\kappa$ -segment phase), is **12 channels** — one PIO state machine each, generating the disk's HV/RF drive timing with no jitter and no CPU in the loop. The fit is one-to-one.
- **Two M33 cores for the two loops.** Core 0 runs the quaternion  $\text{SE}(3)$  mixer (the  $M^{-1}$  solve + frame transforms, already quaternion-native in the SAE control scheme) and the per-disk thrust $\rightarrow$ drive map; Core 1 runs the attitude/rate controller, the EKF (IMU/baro/flow), and the disk health/coherence monitor.
- **Firmware reuse — and it is the same machine.** The output stage is the `12_PORT_SUBSTRATE_ADAPTER` the optical  $x_v$  chambers use. The optical chamber is a 12-port  $x_v$  drive; a Hexalock is six  $x_v$  drives on a flying frame. The coordinator, phase scorebook, and resonance-locking warm loop transfer unchanged; below the adapter the per-port stage drives a disk's HV gate instead of a laser current.
- **Determinism & cost.** A  $\sim \$1$  MCU owns the timing in hardware (PIO), not on a contended RTOS.

RP2350

Core 0

6DOF mixer:  $P[1..6] = M^{-1} \cdot [F_x F_y F_z M_x M_y M_z]^T$  (quaternion)  
per-disk map:  $P_i \rightarrow (|E|_i \text{ drive amplitude}, \kappa_i \text{ asymmetry/sign})$

Core 1

EKF (IMU+baro+optical-flow)  $\rightarrow$  attitude  $q$ , rates  $\omega$ , pos; outer  
attitude/rate PID  $\rightarrow$  wrench  $[F;M]$ ; coherence/HV health; failsafe

PI00 SM0..3  $\rightarrow$  disk 1,2,3,4 drive-amplitude gates (to HV/RF stage)  
PI01 SM0..1  $\rightarrow$  disk 5,6 drive-amplitude gates  
SM2..3  $\rightarrow$  disk 1,2  $\kappa$ -steer: rim-segment phase, or spiral-bias (Arch. C)  
PI02 SM0..3  $\rightarrow$  disk 3,4,5,6  $\kappa$ -steer: rim-segment phase, or spiral-bias + spin-sync  
HW I2C/SPI  $\rightarrow$  IMU (ICM-42688-P), baro (BMP390), mag, flow/ToF  
UART  $\rightarrow$  RX (ELRS/CRSF), telemetry / inter-craft mesh  
ADC/PWM  $\rightarrow$  per-disk field/HV sense; docking-face drivers; LED/buzzer

*One MCU owns all twelve drive channels deterministically. PIO is the low-voltage timing layer; the kilovolt drive lives in a per-disk HV/RF stage (the PIO drives its gate), exactly as the hoverbike's PIO drives class-D amps. Six disks  $\times$  {amplitude,  $\kappa$ } = the 12 PIO state machines.*

### 5.3 Airframe

Keep the prototype's proven structure. A **central octahedral core** (3D-printed centerpiece + six corners, tied by 1.5 mm CFRP triangles) carries the RP2350 board, the power core, and the IMU at the geometric centre; the **outer dodecahedral shell** is CFRP tube and printed pentagon connectors, with the six disks seated on the six pentagon faces. The octahedron-in-dodecahedron is a real packaging win (a rigid central pyramid network) and a nested Platonic solid — the avionics live in the dual of the shell. The disks need a clear, undisturbed face to the outside (their coupling surface), so unlike a ducted rotor there is nothing to shroud and no airflow to manage — the shell can be closed and smooth, which is also what the coherence story of §8 wants.

### 5.4 Sensing & control

The estimator is conventional (IMU + baro + optical-flow/ToF indoors; GNSS + mag outdoors). The human interface follows the hoverbike's "one input device" doctrine, because a 6DOF craft has no inherent "up" to fall back on:

- **Autonomous / pose-hold (default).** The craft holds a pose in  $SE(3)$  at  $\approx$ zero power (§4); the operator commands waypoints and look-directions. Six-DOF station-keeping is an autopilot capability, not a stick skill — and here it is also nearly free to sustain.
- **Two-stick decoupled.** Left stick = body translation (**F**), right stick = body rotation (**M**) — literally the two halves of  $M^{-1}$ .
- **VR / first-person.** The SAE paper's own recommendation: head-tracked goggles with the thrust axes aligned to the pilot's view, so "push forward" is always "where I'm looking." The RP2350 needs only the head quaternion on a UART.

## 5.5 Docking & swarm — the coupling faces

The Lynchpin's symmetric pentagon faces let units **dock in flight** and act as one: the SAE design uses face connectors with spring-dampers (soft capture) and pogo pins (a drone-to-drone bus for comms and power), and a *master-aircraft* scheme where, on connection, one craft solves the others' thrust via a dynamically enlarged mixing matrix ( $N$  docked Hexalocks  $\rightarrow$  one  $6N$ -thrust vehicle). In our frame the docking faces and the lift faces are the *same surface* — the  $x_v$  coupling face (§8) — so bonding two craft is also widening a coherent boundary. Four units close into a gripper (Config A) or a sphere that catches a payload (Config B); that sphere is, not incidentally, a step toward the closed coherent shell of §8.

## §6 Mass, power, and the endurance inversion

A Hexalock's budget is unlike a multirotor's, because nothing converts power into a continuous downward air-flux. Numbers are *ESTIMATES* for sizing, conditional on the disk (§9); they assume a thin diamond/SiC-class disk at the §13 self-lift coupling.

| Component                                                        | Mass                                    | Notes                                                                                         |
|------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------|
| 6× hover disks (thin diamond/SiC core + graphene rim + register) | $\sim 6 \times 60 = 360$ g              | self-lift is per-weight & size-independent ( $F/W \propto R$ ); thinner is easier, not harder |
| 6× per-disk HV/RF drive stages                                   | $\sim 6 \times 35 = 210$ g              | stores $U_{\text{drive}}$ ; the heaviest electronics on the craft                             |
| Octahedral core + dodeca shell (CFRP)                            | $\sim 150$ g                            | closed, smooth — no ducts, no airflow paths                                                   |
| RP2350 board + wiring + sensors                                  | $\sim 40$ g                             | whole flight computer in a palm                                                               |
| Power core (battery / supercap; or fuel cell / rectenna option)  | $\sim 150$ g                            | sized for <i>maneuver work</i> + <i>losses</i> , not for hover                                |
| Docking faces + connectors (swarm units)                         | $\sim 60$ g                             | optional                                                                                      |
| <b>Total (solo)</b>                                              | <b><math>\sim 850</math>–<br/>950 g</b> | comparable to the propeller version — but the power story is inverted                         |

For the **Architecture-C first build** (§5.1), swap the diamond skins for copper shells and add six small spin motors + drivers ( $\sim 6 \times 25 = 150$  g): the solo craft lands  $\sim 1.0$ – $1.1$  kg. The materials-ideal disks later remove the motors and that mass with them. Either way the budget is dominated by *drive electronics*, not by lift hardware — there is no continuous-lift power plant to carry, because the craft *locks* rather than pumps.

**Derived / established · The endurance inversion — lock, don't lift (warp §2).** Holding a hover does no work — the craft **locks**, the lattice holding it the way a table holds a book ( $P = F \cdot v = 0$ ), so a pose-hold draws only the drive field's *losses* (a low-loss / superconducting-boundary disk approaches free hover). Travel costs work, but a conservative coupling is *regenerative*: energy spent accelerating out is recovered decelerating in (KERS-style), so a round trip costs only losses plus net work — not the gross momentum changes. So unlike a multirotor that burns its whole pack just to stay up, a Hexalock's battery is sized for *maneuvering*, and station-keeping endurance is set by losses, not by lift power. (*This is the lawful claim; "free energy / never runs down" is not — you recover what you put in, minus losses.*  
*warp\_drive.html §2.*)

## §7 Things that will go wrong

Field guide, in the hoverbike's spirit — the failure modes of a moving-part-free  $x_v$  6DOF craft:

1. **The disk may not lift at all — that is the headline risk.** Every  $x_v$  measurement to date is null. Build and fly the *single self-lifting disk* on the `warp_drive.html` §7 force balance before you assemble six into an airframe. This vehicle is downstream of that receipt; do not invert the order.
2.  **$\kappa$ -steer calibration is the new mixer input.** The map from commanded thrust  $P_i$  to (amplitude,  $\kappa$ ) per disk must be characterized per disk on a balance, because the §3 mixer assumes a clean, signed, linear actuator. A nonlinear or hysteretic  $\kappa \rightarrow$  force curve re-introduces the very problem collective pitch was trying to remove — table-ize it.
3. **HV breakdown and field health.** The disks store kilovolt fields; breakdown, leakage, or aging shift  $U_{\text{drive}}$  and thus thrust. The Core-1 health monitor must watch each disk's field and fail a disk safely (the §3 mixer can fly degraded on five, with reduced authority).
4. **The simple disk spins — balance it.** Architecture C's constant-RPM flywheel must be balanced (vibration decoheres the shell and corrupts the IMU), and its gyroscopic moment folded into the attitude model; six co-rotating discs add net angular momentum, so alternate spin directions across the six (or null it in the estimator). The materials-ideal static disk has none of this — the spin is the price of the cheap first build, and the reason to climb off it once the bench gives a receipt.
5. **Coherence loss looks like thrust loss.**  $S_{\text{coh}}$  (the coherent fraction) is temperature- and drive-sensitive; a disk that decoheres weakens silently. Treat thermal management of the coherent core as flight-critical, the way a multirotor treats motor temperature.
6. **CG-z still bites.** A vertical CG offset couples into the non-coplanar thrust solution (§3). Keep mass centered in the octahedral core; re-calibrate  $M$  after any payload change.
7. **Disorientation before the craft.** A vehicle with no inherent "up" is disorienting line-of-sight. Fly pose-hold or VR first; two-stick manual is an expert mode.

8. **Docking is a contact-dynamics problem.** Two 6DOF craft capturing face-to-face is where the spring-dampers and a soft-capture state machine earn their mass; dock slow, with the master pre-chosen, and bench the matrix-handover before flying it.

§8 One airframe, one dial — drone to warp-craft

Because the thruster is already a  $x_v$  drive, this airframe does not need an "upgrade path" to reach the warp program — it is the warp program's first vehicle, sitting at the low end of the one coherence dial of `warp_drive.html` §13. The same six (→twelve) faces, driven harder and more coherently, walk the same shell up the ladder:

| Rung               | Coupling<br>$S_{coh}f_{cons}$ | What the Hexalock is                                                                                                                           | Status                                              |
|--------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| lock<br>(hover)    | $\sim 1.4 \times 10^{-6}$     | this drone — six partly-coherent disks, hover & gentle flight, 0.1–1 g                                                                         | OPEN (the bench result this paper is downstream of) |
| flight<br>envelope | $\sim 10^{-3}$                | the Tic-Tac envelope — partial coherent shell, $\sim 700$ g, fast 6DOF                                                                         | OPEN — same disks, stronger drive                   |
| closed<br>shell    | → 0.873                       | the full dodecahedron driven as one coherent shell (the "expanded" Hexalock; or a Config-B swarm sphere) — near-c, then the warp <i>closes</i> | HIGH-GATE — warp §8 (FTL bubble)                    |

Two design facts of the Hexalock make it unusually well-placed on this dial, by virtue of being this shape: it is a *closed, maximally-symmetric dodecahedral shell* (the wholeness/life ether  $S_{coh}$  wants a closed body; the consonance/chemical ether  $f_{cons}$  wants exactly the {5,3,3} cell), and its faces are already *independently driven with controllable asymmetry* (the  $\kappa$  the force direction needs). The "collapsed" six-pentagon form is the drone; the full twelve-face dodecahedron — or four craft bonded into a Config-B sphere — is the closed coherent shell of the starship limit. Howard's swarm bonding is, in this frame, a coherence-area multiplier: docked shells share a wider boundary, raising  $S_{coh}$  by aggregation.

**Doctrine sentence** · One airframe, one dial. Six hover disks in a dodecahedron is a drone at one part in a million of full coherence; the same dodecahedron brought to whole-shell coherence is a warp-craft. The Hexalock does not graduate from drone to starship by changing shape — it climbs the coherence-and-consonance curve of the warp drive while staying the exact same cell. Build the drone; you have built the first rung of the starship.

§9 What is established, and what is open



| Element                                                                                                                               | Status                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Six-pentagon collapsed-dodecahedron geometry; six lin.-indep. thrust vectors; $M$ square & invertible → 6DOF                          | ESTABLISHED (SAE 2023, derived & flight-tested)                                                                                   |
| The 6DOF mixer / quaternion control flies with real vectored thrusters                                                                | ESTABLISHED — the propeller predecessor flew; the mixer is thruster-agnostic                                                      |
| A $x_v$ disk gives signed, latency-free, moving-part-free, air-free thrust, and $\approx$ free hover                                  | DERIVED from the force law & warp §2 — <i>conditional</i> on the disk lifting at all                                              |
| RP2350: 12 PIO SMs = 6 disks $\times$ {amplitude, $\kappa$ }; dual-M33 mixer/estimator; firmware reuse from the 12-port $x_v$ adapter | ESTABLISHED capability; the mapping is THIS PAPER'S DESIGN                                                                        |
| Airframe = the {5,3,3} lattice cell; disks lift by resonating with that cell                                                          | ESTABLISHED geometry; the lattice identification is the network ontology                                                          |
| <b>The hover disk's lift magnitude</b> — does a real disk self-lift?                                                                  | OPEN — force law derived in <i>form</i> ; magnitude unmeasured, null to date. The whole vehicle is downstream of this one number. |
| The flight / Tic-Tac envelope and the closed-shell warp limit                                                                         | OPEN / HIGH-GATE — warp_drive.html §8                                                                                             |

**Open — to be confirmed on the bench** · The honest one line: **this is the vehicle the hover disk delivers, not a vehicle that flies today**. The geometry, the flight-proven 6DOF mixing, and the RP2350 control are sound and buildable now; the lift is the single open bench question of the companion papers. The right order is the `hover_disk.html` tier ladder — first the self-lifting disk on the `warp_drive.html` §7 force balance, then six of them in this airframe. We never let the Lynchpin's flight-proven control half lend its certainty to the  $x_v$  lift half; they are two ledgers, and only one is closed.

## §10 Conclusion

Put a propeller in each pentagon and the airframe is a clever omnidirectional drone. Put a hover disk in each pentagon and it is something else: a propellantless, silent, moving-part-free craft that hovers at any attitude in any medium for almost no power, mixed and driven by a single RP2350 whose twelve channels fall naturally onto the six disks' drive. The airframe's 6DOF control is flight-proven; the disk's lift is the open bench question of the hover/warp papers — and the contribution of this paper is to show that the geometry, the mixing, and the controller are ready and waiting, so that the day a single disk self-lifts, six of them are a craft. It is the same dodecahedral cell all the way up the

coherence dial: a drone at a millionth of full coherence, a starship at the whole shell. Build the self-lifting disk first. Then build six, and fly the cell.

## References

1. Howard, T.D., Molter, C., Seely, C.D. & Yee, J. (2023). "The Lynchpin — A Novel Geometry for Modular, Tangential, Omnidirectional Flight." *SAE Int. J. Aerosp.* 16(3), Article 01-16-03-0018, doi:10.4271/01-16-03-0018 . (Geometry, the six-thrust-vector mixing matrix, the flight-tested 6DOF prototype and quaternion control, the docking/swarm configurations. Used here for the geometry and the flight-proven control; the propeller propulsion is replaced.)
2. *The Hover Disk*, [hover\\_disk.html](#) — the  $x_v$  lift body that fills each pentagon: the substrate ladder (diamond/SiC/graphene), the four ethers, the rim-segmented drive and capacitor architecture, and the honest derived/open status the disk's lift carries.
3. Viktor Schauburger — *The Water Wizard and the Inward Drive*, [viktor\\_schauburger.html](#) §9 — "Architecture C, the spun charged chiral-vortex shell" (Repulsive + Tesla bladeless turbine + resonant coil): the practically-simple first-disk build of §5.1, a bench hypothesis exercising four warp-drive levers the static capacitor does not. Right shape; the lift is the open bench question.
4. *The Warp Drive*, [warp\\_drive.html](#) — the propellantless lattice-drive mechanism; §3 the force-as-pressure / self-lift law, §2 why hover is  $\approx$ free and travel is regenerative, §13 the closed-form thrust and the one coherence dial (self-lift  $\sim 10^{-6} \leftrightarrow$  FTL bubble).
5. *The DIY Hoverbike*, [diy\\_hoverbike\\_paper.html](#) — the RP2350 / 12-port substrate adapter / "one input device" engineering-optimum template this build follows.
6. *How to Build a UFO*, [how\\_to\\_build\\_a\\_ufo.html](#) §6/§10 — the closed-shell, rim-segmented saucer morphology each disk and the whole shell reuse.
7. *The Shape of the Universe*, [shape\\_of\\_the\\_universe.html](#) ; *The Periodic Table*, [the\\_periodic\\_table.html](#) §13 — the {5,3,3} dodecahedral lattice and the dodecahedrane  $C_{20}$  primitive cell the airframe instantiates. Vertex-sharing bound: [G\\_VERTEX\\_DERIVATION.md](#) ( $g \leq 1/\sqrt{3}$ ); force law: [CHI\\_NU\\_FORCE\\_LAW.md](#) .
8. Brescianini, D. & D'Andrea, R. — the ETH "Omnicopter," an eight-rotor 6DOF vehicle; Ryll, M. (2015), overactuated quadrotor (University of Stuttgart). (The 6-thruster near-zero-thrust constraint of §3, which the  $x_v$  disk dissolves.)
9. Raspberry Pi Ltd. — *RP2350 Datasheet* (dual Cortex-M33 + dual RISC-V Hazard3; three PIO blocks / twelve state machines). The §5 controller.

Draft engineering paper in the  $x_v$  / vertex-sharing research program. It specifies a propellantless 6DOF craft that places a hover disk (the  $x_v$  lift body of [hover\\_disk.html](#) ) in each pentagon of the flight-proven Lynchpin geometry (Howard et al., SAE 2023). The geometry and the 6DOF mixing/control are established and flight-proven

with a propeller predecessor; the **lift itself is the open bench question** of the companion papers — the force law is derived in *form*, its magnitude unmeasured and null to date — and the whole vehicle is downstream of that single measurement ([warp\\_drive.html](#) §7). Quantities are tagged `ESTABLISHED` / `DERIVED` / `ESTIMATE` / `OPEN`. The Lynchpin geometry and "tangential flight" are trademarks of Tangential Flight Corporation; this paper analyzes and builds on the published design and claims no affiliation.